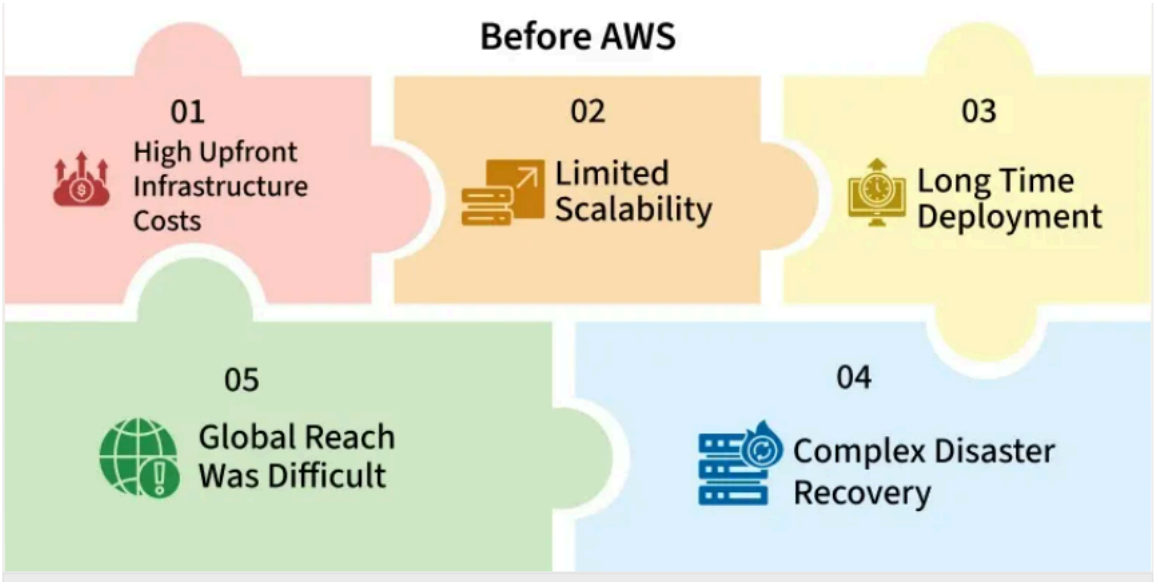
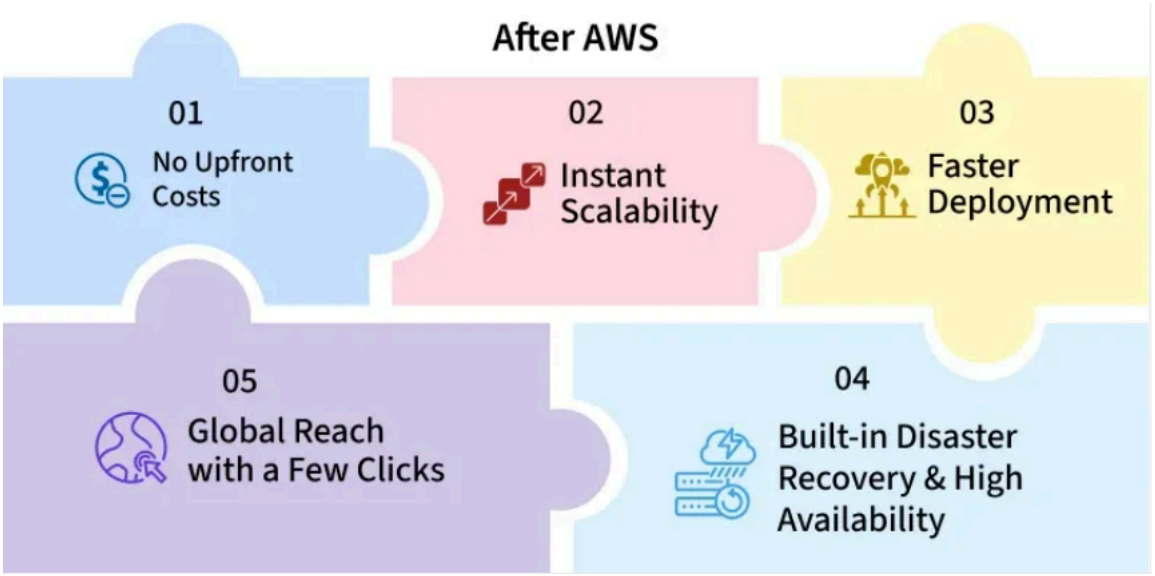
Syllabus (Module 1 Part 1):- AWS Cloud Foundations: Introduction to Cloud Computing, Overview of AWS Global Infrastructure

Refer-> https://aws.amazon.com/documentation-overview/

Amazon Cloud Services, also known as Amazon Web Services (AWS), is a cloud computing platform where Amazon provides a range of information technology services including servers, storage, databases, networking, artificial intelligence, analytics, and more. AWS allows organizations and individuals to utilize the infrastructure and services they need in a flexible manner.

Amazon Web Services (AWS) was started in 2006 to help companies avoid the high cost and effort of buying and managing their servers. Before AWS, businesses had to set up physical computers and storage to run websites or apps, which took time and money. AWS came into the market to solve this problem by offering these resources over the internet.





How AWS Delivers Cloud Services

Amazon Web Services (AWS) delivers cloud computing through a network of globally distributed

data centers connected by high-speed fiber. These centers host a wide range of services from storage and compute power to databases and machine learning that users can access on demand via the internet.
Whether you are a startup or an enterprise, AWS allows you to build and run applications without managing physical servers. AWS handles:

- Server maintenance
- Security patching
- Resource scaling
- High availability.

For example, if you are running an eCommerce site, instead of manually handling database setup, backups, and failover, you can use **Amazon RDS**. It automates all these backend tasks, enabling you to launch a database in minutes with just a few clicks.

Some key features and services of AWS include:
1. Elastic Infrastructure: AWS enables users to scale their resources up or down based on demand. This is particularly advantageous for websites or applications with variable traffic loads.

2. Wide Service Portfolio: AWS offers a diverse range of services in computing, storage, databases, artificial intelligence, analytics, networking, security, IoT, and more.

3. Scalability: AWS is designed to support large-scale projects and applications. Resources can be easily scaled and expanded.

4. Payment Model: AWS allows users to pay for the resources and services they use, which can help optimize costs.
How is AWS able to offer pay-as-you-go?
AWS separates different virtual machines from each other on the same hardware. So for example, my company and Ravina's company could be sharing the exact same physical server owned by AWS, but our virtual machines would still be completely separated. This ensures that servers are put to their highest and best use, allowing cheaper server costs to us as users than what we'd typically have to pay for to manage our own data server (unless my company becomes as big as Amazon).



5. Rapid Deployment: AWS facilitates quick deployment and updating of applications and services.

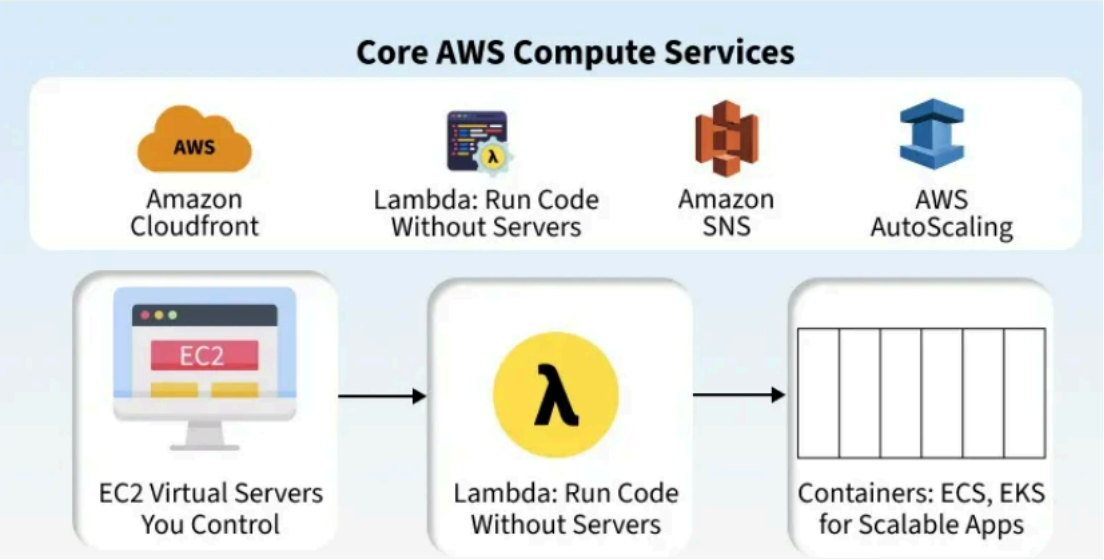
6. High Security: AWS has high-security measures and services to protect user data.

7. Global Infrastructure: AWS has numerous data centers and regions located worldwide, ensuring high availability and low latency.

8. Ease of Management: AWS provides a user-friendly management console to help users manage and monitor resources.
Amazon Cloud Services is a comprehensive platform catering to data storage, computational power, application deployment, database management, and more for businesses, developers, and individuals of varying sizes.

Here are some real-world use cases of AWS services:

- **Netflix:** The large streaming giant using AWS for the storage and scaling of the applications for ensuring seamless content delivery with low latency without interruptions to millions of users globally.
- **Airbnb:** By utilizing AWS, Airbnb manages the various workloads and provides scalable and reliable infrastructure for its virtual marketplace and lodging offerings.
- **NASA's Jet Propulsion Laboratory:** It takes the help of AWS services to handle and analyze large-scale volumes of data related to vital scientific research missions and space exploration.



Cloud computing is a computing paradigm that enables on-demand access to a shared pool of configurable computing resources (such as servers, storage, databases, networking, software, and analytics) over the internet. It allows users to provision and release resources with minimal management effort or service provider interaction, making it highly scalable and cost-effective for organizations.

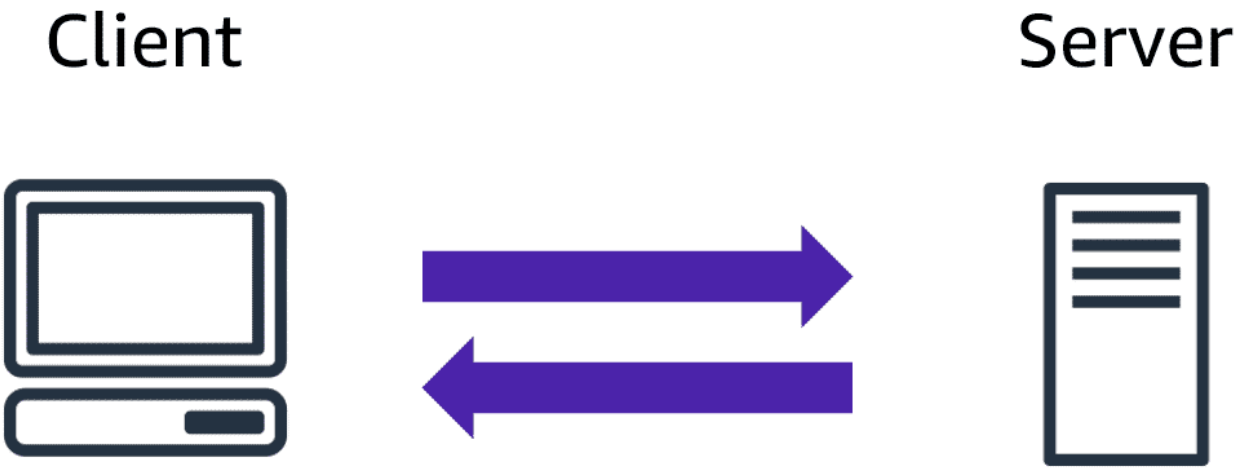
Definition: Delivery of computing services over the internet, billed based on usage similar to utilities like water or electricity.

The Client-Server Model

- The client-server model is an important concept in cloud computing.

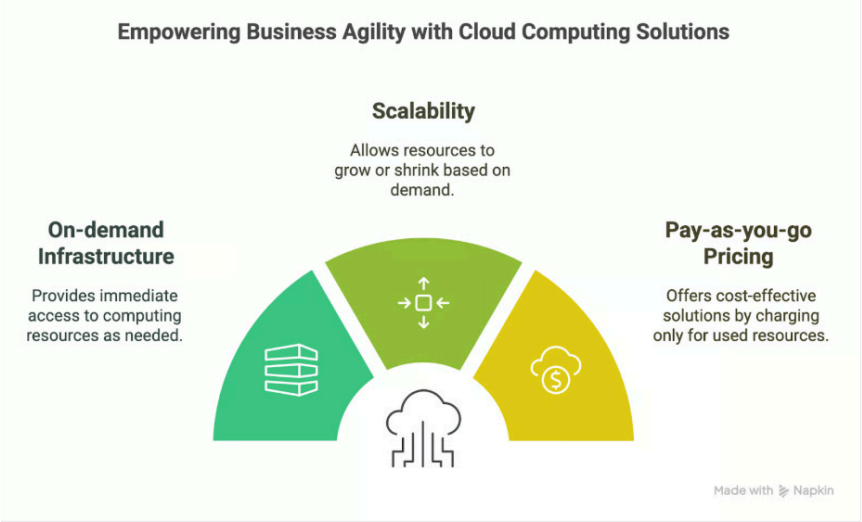
- It is about many clients using services from a centralized server.

The Client-Server model is about a client that interacts and makes requests to a computer server. A client is the way that the person interact with the server.



The server does tasks for the client and returns information.

- Delivery Models:
  - Infrastructure as a Service (IaaS): Virtualized computing resources online. (refer :- > https://aws.amazon.com/what-is/iaas/)



A diagram showing the key features of Infrastructure as a Service (IaaS).

The following are some of the key features of IaaS:

- **On-demand infrastructure**: You can scale resources like storage, computing power, and bandwidth without needing to purchase additional hardware.
- **Scalability**: IaaS providers offer flexible scaling options, which allow you to adjust resources based on current needs.
- **Pay-as-you-go pricing**: IaaS typically uses a pay-per-use model, which helps businesses save on operational costs by only paying for the resources they use.

Common use cases

The following are some of the common use cases of IaaS:

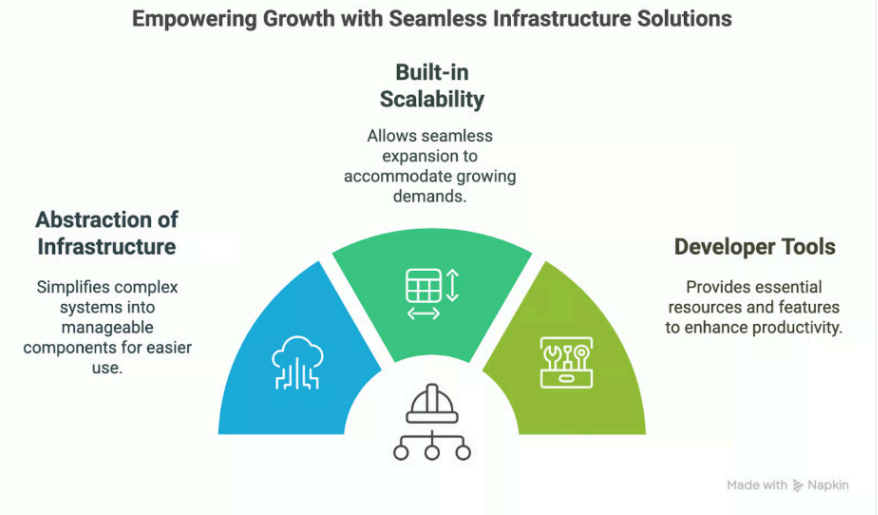
- **Hosting websites**: IaaS is commonly used to host websites, which provides both the computing resources and network capabilities to handle large amounts of traffic.
- **Disaster recovery**: Many businesses use IaaS for backup and disaster recovery solutions, which offer high availability without requiring substantial investments in physical hardware.
- **Test environments**: Developers can use IaaS to create test environments and quickly spin up virtual machines to test applications without affecting live production environments.

Popular IaaS providers

The following are some of the popular IaaS providers:

- **AWS EC2**: Amazon Web Services (AWS) Elastic Compute Cloud (EC2) is one of the most popular IaaS offerings, providing scalable compute capacity in the cloud.
- **Google Compute Engine**: Google's IaaS offering, which provides virtual machines and cloud storage.
- **Microsoft Azure VMs**: Microsoft Azure's virtual machine service enables you to run virtual servers and scale based on demand.

- Platform as a Service (PaaS): Cloud-based app development platforms.
(refer :- > https://aws.amazon.com/what-is/ipaas/)



A diagram showing the key features of Platform as a Service (PaaS).

The following are some of the key features of PaaS:

- **Abstraction of infrastructure**: PaaS abstracts most of the infrastructure management, so developers can focus on application development instead of dealing with the complexities of servers and storage.
- **Built-in scalability**: PaaS platforms often include automatic scaling capabilities, so applications can grow based on demand.
- **Developer tools**: PaaS services typically come with built-in tools for application development, including APIs, databases, and frameworks, streamlining the development process.

Common use cases

The following are some of the common use cases of PaaS:

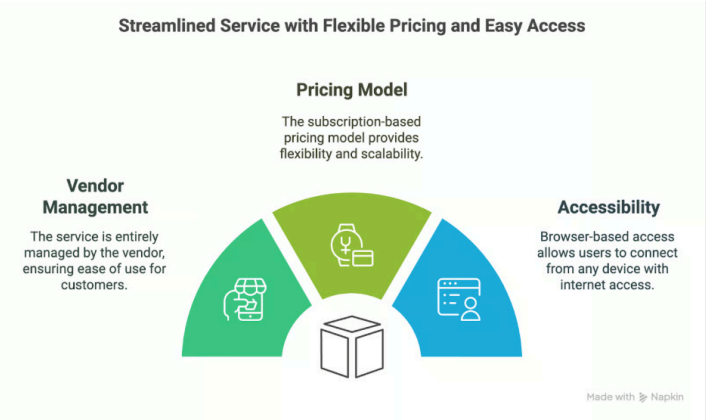
- **Web app development**: PaaS is commonly used to develop and deploy web applications with minimal infrastructure management.
- **API management**: Developers can use PaaS to manage and deploy APIs, ensuring seamless communication between different services and applications.
- **Microservices architecture**: PaaS is often used to build microservices, which are small, independently deployable units of functionality that can be scaled individually.

Popular PaaS providers

The following are some of the popular PaaS providers:

- **Heroku**: A popular PaaS offering that allows developers to build and deploy web applications without worrying about infrastructure.
- **Google App Engine**: Google's PaaS platform provides an environment for building and deploying applications in various languages, including Python, Java, and Go.
- **AWS Elastic Beanstalk**: Amazon's PaaS offering, which allows developers to deploy and manage applications without dealing with the underlying infrastructure.

- Software as a Service (SaaS): Web-based applications accessible via browsers. (refer :- > https://aws.amazon.com/what-is/saas/)



A diagram showing the key features of Software as a Service (SaaS).

The following are some of the key features of SaaS:

- **Fully managed by the vendor**: The service provider takes care of all updates, security patches, and maintenance.
- **Subscription-based pricing**: SaaS is usually offered through subscription-based pricing models, making it cost-effective for businesses of all sizes.
- **Browser-based access**: You access SaaS applications via a web browser, removing the need for local installations or hardware requirements.

Common use cases

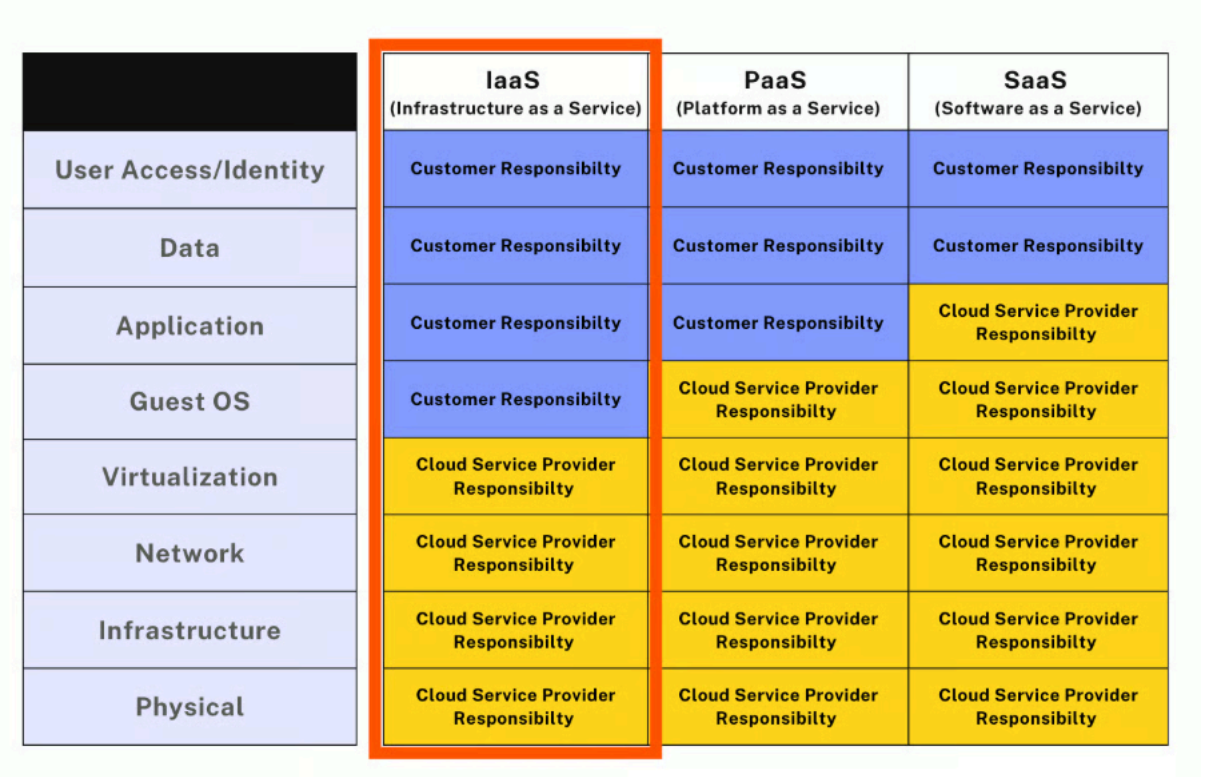
The following are some of the common use cases of SaaS:

- **Email**: Services like Gmail or Outlook are classic examples of SaaS for email communication.
- **Customer Relationship Management (CRM)**: SaaS platforms like Salesforce help businesses manage customer relationships, track sales, and analyze data.
- **Project management**: SaaS tools like Trello and Asana provide collaborative platforms for managing tasks and projects.

Popular SaaS examples

The following are some of the popular SaaS examples:

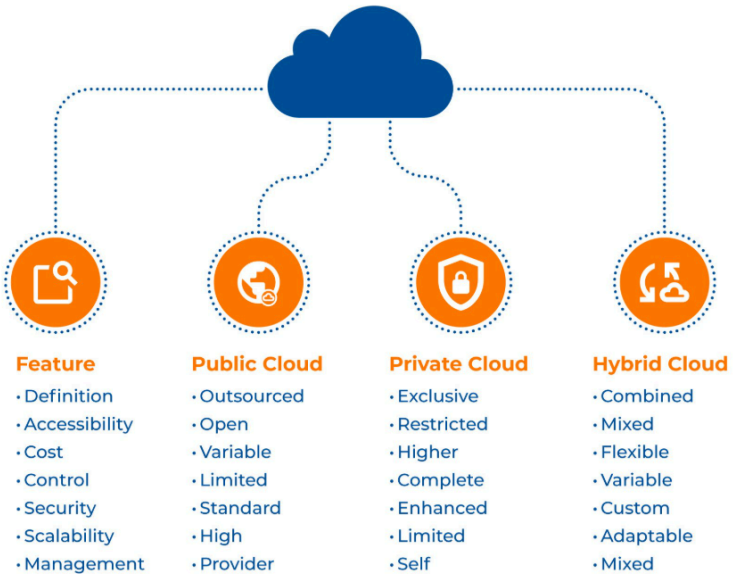
- **Salesforce**: A leading SaaS solution for CRM, helping businesses manage their customer relationships and sales pipelines.
- **Google Workspace**: A suite of productivity tools (Docs, Sheets, Drive, etc.) delivered as SaaS.
- **Microsoft 365**: Microsoft's cloud-based suite for word processing, spreadsheets, email, and collaboration.



	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
User Access/Identity	Customer Responsibilty	Customer Responsibilty	Customer Responsibilty
Data	Customer Responsibilty	Customer Responsibilty	Customer Responsibilty
Application	Customer Responsibilty	Customer Responsibilty	Cloud Service Provider Responsibilty
Guest OS	Customer Responsibilty	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty
Virtualization	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty
Network	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty
Infrastructure	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty
Physical	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty	Cloud Service Provider Responsibilty

- Deployment Models:
  - Public cloud: Accessible via the internet, provided by third parties.
  - Private cloud: Exclusively operated for a single organization.
  - Hybrid cloud: Mix of private and public cloud resources.

  (Refer- > https://aws.amazon.com/what-is/private-cloud/)
  (Refer -> https://www.geeksforgeeks.org/computer-science-fundamentals/cloud-deployment-models/)



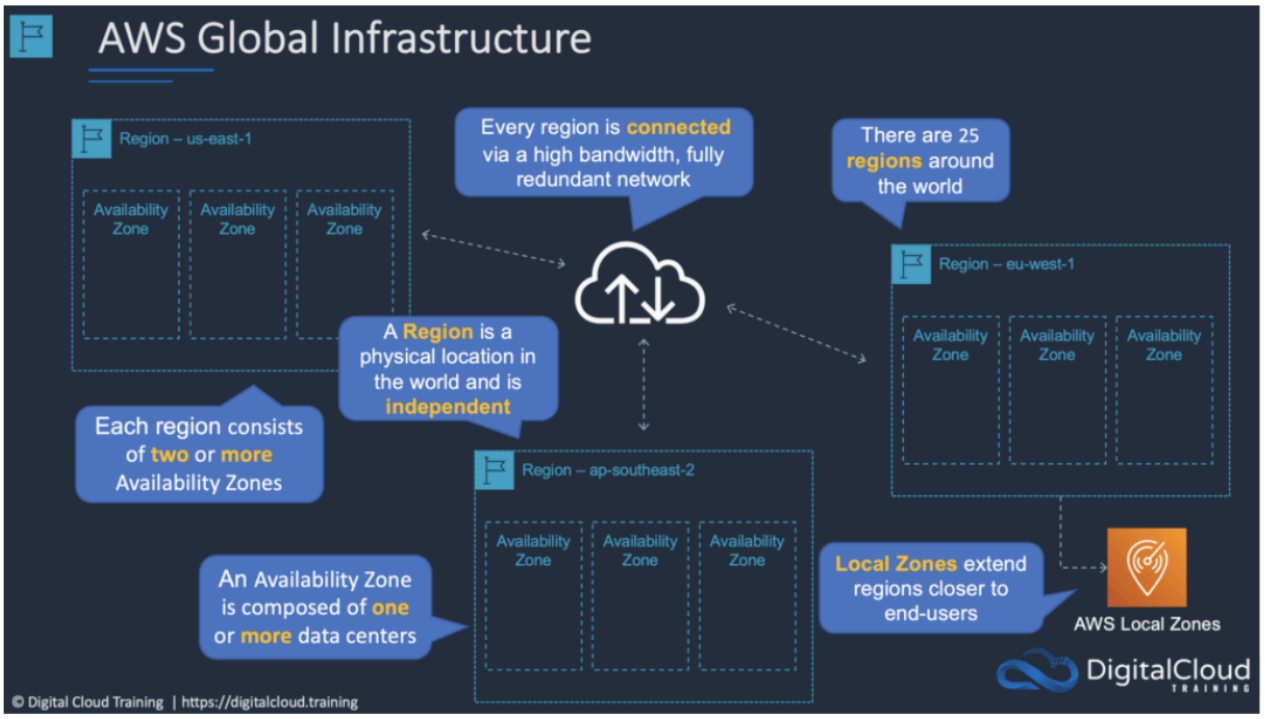
Overview of AWS Global Infrastructure

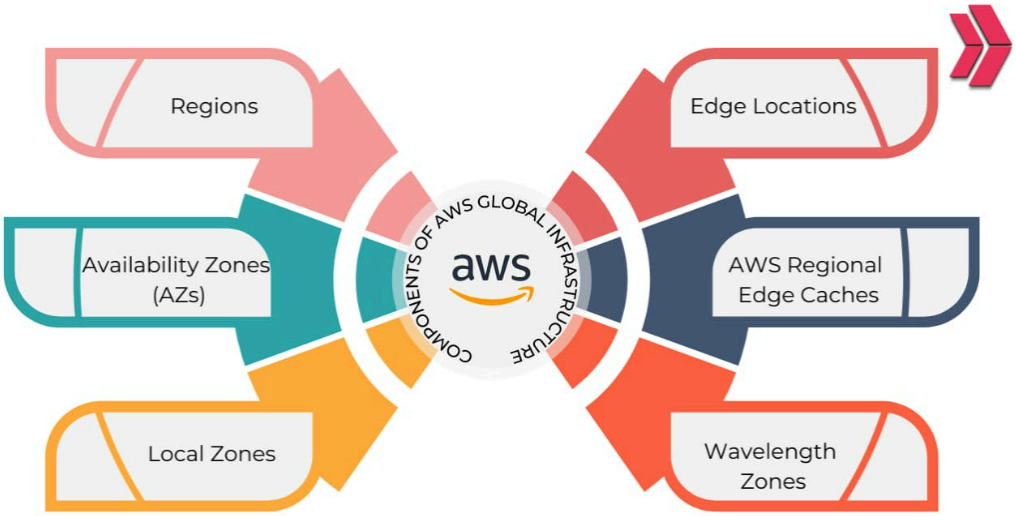
Amazon Web Services (AWS) operates the largest cloud computing infrastructure in the world, composed of multiple interconnected components that deliver reliable, scalable services globally.

The AWS Global Infrastructure is a vast network of data centers and resources strategically located throughout the world to provide cloud services. The infrastructure comprises multiple interconnected components, such as Regions, Availability Zones, Local Zones, and Wavelength Zones.

AWS Global Infrastructure's main component is data centers called Availability Zones (AZ). AZ's are isolated locations within AWS Regions. The AWS Global Infrastructure Map currently includes **102 Availability Zones** spread across **32 geographic regions** globally. Moreover, plans are underway to expand with an additional **12 Availability Zones** and **4 more AWS Regions**, covering countries like Canada, Malaysia, New Zealand, and Thailand.

Additionally, AWS offers **35 Local Zones** and **29 Wavelength Zones**, catering to the needs of ultralow latency applications. Local Zones extend AWS services to select metropolitan areas, allowing for proximity to end-users and reduced latency. On the other hand, telecommunication providers collaborate in designing Wavelength Zones to bring AWS services even closer to 5G networks, thereby enhancing the performance of latency-sensitive applications.





Core Components

- Regions: Physical locations worldwide where AWS clusters data centers. Each region is independent and contains multiple Availability Zones.
- Availability Zones (AZs): Groupings of discrete data centers within each region with redundant power, networking, and connectivity for fault tolerance and high availability.
- Edge Locations: Over 400 sites used by services like Amazon CloudFront to deliver content with low latency globally.
- Local Zones & Wavelength Zones: Provide compute and storage services closer to users or within telecom networks for ultra-low latency applications (like gaming or real-time analytics).
- Outposts: Extend AWS infrastructure on-premises, enabling hybrid cloud solutions.

AWS Regions

An AWS Region is a distinct geographical area worldwide where a collection of data centers is clustered. This model incorporates multiple Availability Zones and physically and logically separate

logical data center groups. Each AWS Region comprises **at least three** such Availability Zones, ensuring geographic redundancy and high availability.
AWS Regions and Availability Zones are not the same. AWS Regions are the larger geographical areas where data centers are located, while Availability Zones are subsets within those regions. We will discuss AZs deeply in the next part.
AWS operates in several geographic regions, including North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

AWS Availability Zones

People often refer to Availability Zones (AZs) in AWS as data center facilities. Each Availability Zone is a physically separated data center with its own power and infrastructure. AWS has strategically designed these Availability Zones to provide redundancy and resiliency for applications and services.
AWS Availability Zones have their own power, cooling, and networking infrastructure. Deploying resources across multiple Availability Zones ensures that if one AZ experiences issues or downtime, it enables the others to continue operating, thereby enhancing the overall resilience of applications hosted on the AWS platform.
Physical distances are important when comparing the distance between an AWS Region and its corresponding AZ. This separation helps minimize the impact of natural disasters or localized incidents. A significant distance physically separates Availability Zones in AWS. It's important to note that while the **distance between AZs** is substantial, all AZs within a given AWS Region remain within **100 kilometers** (approximately 60 miles) from each other.
Here is the full list of AWS data center locations in all regions:
Europe / Middle East / Africa

- **11 Geographic Regions**
- **33 Availability Zones**
- **39 Edge Network locations**
- **2 Regional Edge Cache locations**

Availability Zones:
Bahrain (**3**), Cape Town (**3**), Frankfurt (**3**), Ireland (**3**), London (**3**), Milan (**3**), Paris (**3**), Spain (**3**), Stockholm (**3**), Zurich (**3**), and UAE (**3**)
North America

- **7 Geographic Regions**
- **25 Availability Zones**
- **44 Edge Network locations**
- **2 Regional Edge Cache locations**

Availability Zones:
N. Virginia (**6**), Ohio (**3**), N. California (**3**), Oregon (**4**), US-East (**3**), US-West (**3**), Central (**3**)
South America

- **1 Geographic Regions**
- **3 Availability Zones** (in São Paulo)
- **4 Edge Network locations**
- **1 Regional Edge Cache locations**

Asia Pacific and China

- **12 Geographic Regions**
- **38 Availability Zones**
- **34 Edge Network locations**
- **5 Regional Edge Cache locations**

AWS Local Zones

AWS Local Zones are a part of the AWS Global infrastructure that brings cloud services locally to certain areas with a high concentration of users and applications. Essentially, these Local Zones form a subset of AWS Availability Zones, but they situate themselves in close proximity to specific metropolitan areas.

Local Zones allow customers to deploy applications that require low latency to end-users or specific resources in those geographic areas. They are particularly useful for applications that require real-time processing, such as gaming, interactive multimedia, and financial services, where reducing latency is critical for a seamless user experience.

AWS Local Zones Locations:

Local Zones are available in **33 metropolitan areas** around the world, where **16 of them** are located in the US (specific names not provided), and **17 of them** are located outside of the US. Here are **17 Local Zones** located outside the US:

1. Lagos
2. Auckland
3. Santiago
4. Bangkok
5. Kolkata
6. Copenhagen
7. Hamburg
8. Helsinki
9. Lima
10. Muscat
11. Perth
12. Querétaro
13. Taipei
14. Warsaw
15. Delhi
16. Buenos Aires
17. Manila

Additionally, AWS plans to broaden Local Zones to encompass **19 additional locations** spanning **16 countries**:

1. Australia
2. Austria
3. Belgium
4. Brazil
5. Canada
6. Colombia
7. Czech Republic
8. Germany
9. Greece
10. India
11. Kenya
12. Netherlands
13. Norway
14. Portugal
15. South Africa
16. Vietnam

AWS Edge Locations

AWS Edge Locations are endpoints for Amazon Web Services' CloudFront service. They are strategically distributed data centers situated in various areas around the world. These Edge Locations act as caching servers that store copies of frequently accessed content, such as images, videos, web pages, and other static files, closer to the end users.

The Edge Location closest to the client receives requests for content distributed via CloudFront. This reduces latency and improves the overall performance of the application or website if the content already exists at that Edge Location. If the content still needs to be cached, the Edge Location retrieves it from the origin server (such as an Amazon S3 bucket or a custom origin) and

caches it for subsequent requests.

AWS Regional Edge Caches

Like edge locations, Regional Edge Caches are CloudFront service sites strategically placed worldwide near your viewers. They deliver content directly to viewers between your origin server and global edge locations.

Regional edge caches assist with many sorts of material, especially with content that becomes less popular over time. Examples are user-generated material, e-commerce assets, news, and other content that may rapidly gain popularity.

Wavelength Zones

AWS Wavelength is an Amazon Web Services (AWS) service that brings cloud services to the edge of 5G networks. It supports real-time gaming, augmented reality (AR), virtual reality (VR), video streaming, and other applications that require ultra-low latency and high bandwidth connectivity.

A Wavelength Zone is essentially a specialized infrastructure deployment within a data center run by a telecommunications provider.

These zones are strategically placed in close proximity to 5G base stations, allowing for a direct connection between the AWS resources and the 5G network. The proximity minimizes latency for applications requiring instant responses.

With Wavelength, developers can run their applications and workloads on AWS infrastructure located at the edge of the 5G network, combining the benefits of both AWS services and high-speed, low-latency 5G connectivity.

Latest Scale (2025)

- 38 Regions
- 120 Availability Zones
- 400+ Edge Locations

Role in Cloud Computing

- Ensures redundancy, resilience, and compliance.
- Regions and AZs enable disaster recovery, data sovereignty, and high availability.



Refer -> https://www.geeksforgeeks.org/cloud-computing/aws-global-infrastructure/

AWS Shared Responsibility Model

Refer -> https://docs.aws.amazon.com/whitepapers/latest/aws-risk-and-compliance/shared-responsibility-model.html

The AWS Shared Responsibility Model defines the division of security and compliance

responsibilities between AWS and its customers.

Principle

- AWS responsibility: Security of the cloud – protecting the infrastructure, including hardware, software, networking, and facilities running AWS services.
- Customer responsibility: Security in the cloud – managing and securing data, applications, and configurations within their own AWS environment.

Examples

- AWS secures the physical infrastructure (data centers, servers, facilities).
- Customers secure operating systems, applications, and data, including setting up firewalls, encryption, identity management, and compliance controls applicable to their use case.
- For Amazon EC2 (IaaS): Customer handles OS, applications, and firewall (security group) configuration.
- For Amazon S3: Customer manages data access policies and encryption methods.

Implications

- Enables customers to focus on securing workloads, while AWS manages the platform's foundational security.
- Responsibilities shift based on the AWS services used (IaaS vs. abstracted services).
- Customers must not rely solely on AWS—ownership of risk management is critical.

What Is AWS Responsible For?

AWS is responsible for security of the cloud. This means AWS is required to protect and secure the infrastructure that runs all the services offered in the AWS Cloud. AWS is responsible for:

- Protecting and securing AWS Regions, Availability Zones, and data centers, down to the physical security of the buildings
- Managing the hardware, software, and networking components that run AWS services, such as the physical server, host operating systems, virtualization layers, and AWS networking components

The level of responsibility AWS has depends on the service. AWS classifies services into three different categories. The following table provides information about each, as well as the AWS responsibility.

Category	Examples of AWS Services in the Category	AWS Responsibility
Infrastructure services	Compute services, such as Amazon Elastic Compute Cloud (Amazon EC2)	AWS manages the underlying infrastructure and foundation services.
Container services	Services that require less management from the customer, such as Amazon Relational Database Service (Amazon RDS)	AWS manages the underlying infrastructure and foundation services, operating system, and application platform.
Abstracted services	Services that require very little management from the customer, such as Amazon Simple Storage Service (Amazon S3)	AWS operates the infrastructure layer, operating system, and platforms, as well as server-side encryption and data protection.

What Is the Customer Responsible For?

You're responsible for security in the cloud. When using any AWS service, you're responsible for properly configuring the service and your applications, as well as ensuring your data is secure.

The level of responsibility you have depends on the AWS service. Some services require you to perform all the necessary security configuration and management tasks, while other more abstracted services require you to only manage the data and control access to your resources. Using the three categories of AWS services, you can determine your level of responsibility for each AWS service you use.

Category	AWS Responsibility	Customer Responsibility
Infrastructure services	AWS manages the infrastructure and foundation services.	You control the operating system and application platform, as well as encrypting, protecting, and managing customer data.
Container services	AWS manages the infrastructure and foundation services, operating system, and application platform.	You are responsible for customer data, encrypting that data, and protecting it through network firewalls and backups.
Abstracted services	AWS operates the infrastructure layer, operating system, and platforms, as well as server-side encryption and data protection.	You are responsible for managing customer data and protecting it through client-side encryption.

Due to the varying level of effort, it's important to consider which AWS service you use and review the level of responsibility required to secure the service. It's also important to review how the shared security model aligns with the security standards in your IT environment, as well as any applicable laws and regulations.
It's important to note that you maintain complete control of your data and are responsible for managing the security related to your content. Here are some examples of your responsibilities in context.

- Choosing a Region for AWS resources in accordance with data sovereignty regulations
- Implementing data protection mechanisms, such as encryption and managing backups
- Using access control to limit who has access to your data and AWS resources

